SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID -** SIH26204
- **Problem Statement Title -** Student Innovation
- **Theme -** Travel & Tourism
- **PS Category -** Software
- **Team ID -** (to be allotted)
- **Team Name –** SafarMitra

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TransitMitra: Offline-First Travel Safety & Mobility Mesh for Network Dead Zones

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Problem:

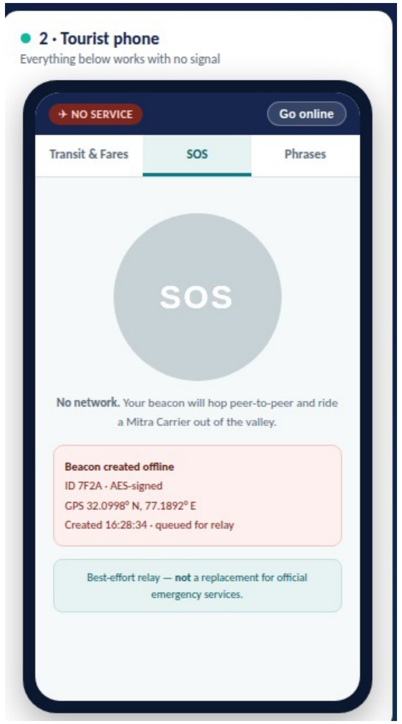
Remote tourist belts suffer a **connectivity blackout** — no network means no maps and no way to call for help; unmapped jeeps and autos **overcharge tourists**; and a language barrier blocks asking locals. Travellers avoid these places, so rural hotels, homestays and transport operators **lose their income.**

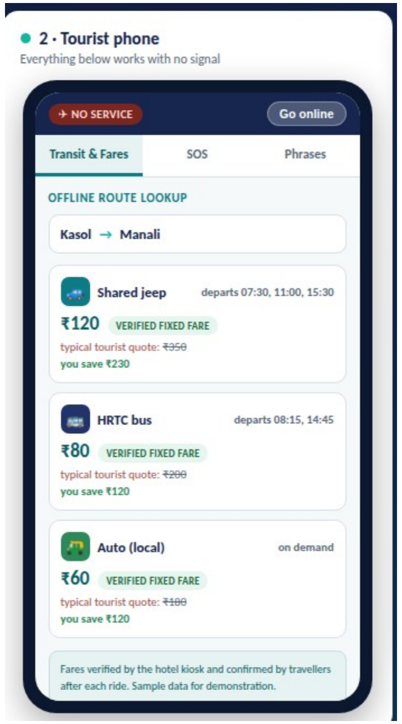
Our Idea :

TransitMitra is an **offline-first travel companion** that works with zero internet, no SIM and no new telecom towers. Hotels become trusted data kiosks, and ordinary buses and jeeps become **moving data carriers** that carry emergency signals out of dead zones.

Proposed Solution :

- Hotel “Mitra Kiosk” sync — on check-in the hotel device pushes the regional transit map, verified fares and emergency directory to the guest phone over Wi-Fi Direct / BLE.
- Verified transit & fares — hotel-verified routes, timings and fixed fares for buses, shared jeeps and autos, ending tourist overcharging.
- Store-and-forward SOS — in dead zones a signed beacon hops phone-to-phone and rides a passing bus until it reaches network, then auto-alerts authorities with GPS.
- Offline language help — regional phrasebook with on-device text-to-speech.





Working prototype — offline fare lookup & SOS

Innovation/Uniqueness:

- “Mitra Carrier” — scheduled buses & jeeps act as data mules, so an SOS leaves the valley on a predictable backbone, not by luck.
- Hospitality becomes the network — hotels are static data nodes; no new telecom infrastructure needed.
- One mesh, three jobs — SOS carrier, live fare/timing source and moving relay.
- Delay-Tolerant design works where every other travel app goes dark.

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TECHNICAL APPROACH

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Technologies Used

- Languages: Dart, Kotlin, Java, Python, C/C++, SQL.
- Mobile: Flutter / Android native (entry-level Android supported).
- P2P transport: Google Nearby Connections, Wi-Fi Aware / Wi-Fi Direct, BLE advertising.
- Long range: LoRa / Meshtastic relay nodes (₹300–500 each, km range).
- Data: SQLite, CRDT conflict-free offline merge, DTN store-and-forward (RFC 5050).
- AI / voice: TensorFlow Lite, on-device text-to-speech for offline phrasebook.
- Security: AES-signed, rate-limited SOS beacons; hotel-verified fare records.
- Prototype: self-contained HTML/CSS/JS demo console, runs fully offline.

FLOW CHART

Store-and-forward SOS relay — beacon escapes the dead zone by physically travelling out

Methodology

- Sync at hotel kiosk → offline data on phone.
- Beacon created offline → BLE peer hop → Mitra Carrier → cellular edge → authorities.
- Pilot 1 circuit → add LoRa relays → scale via State Tourism departments.

GitHub link: github.com/<your-username>/transitmitra
Live demo :
<your-username>.github.io/transitmitra
YouTube video :
Link
Above 60% of the prototype is completed

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FEASIBILITY AND VIABILITY

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Feasibility

- Runs on the cheap entry-level Android phones tourists already carry.
- Uses free peer-to-peer radios already in every phone — no new telecom towers, no spectrum licence.
- Range extended by LoRa relay nodes costing only ₹300–500 each.
- Hotels and homestays adopt the kiosk free of cost, giving ready distribution.
- Proven research base: Delay-Tolerant Networking, MIT DakNet, KioskNet.

Commercial Feasibility

- Market: rural and offbeat tourism is a fast-growing segment currently held back by network dead zones.
- Near-zero running cost — the network is the phones travellers already own.
- Revenue path: hotel subscriptions, State Tourism-department sponsorship of relay nodes.
- Technical readiness: working prototype today; 0–3 month pilot on a single tourist circuit.
- Scales to any off-grid belt by adding regional map and fare packs.

Challenges

- Sparse phone density — an SOS may wait for a carrier to pass.
- Short BLE / Wi-Fi range in mountainous terrain.
- Fake fare entries or spoofed SOS beacons.
- Battery drain from continuous radio scanning.
- Keeping offline data trustworthy across devices.

Strategy

- Register buses and jeeps as scheduled “Mitra Carriers” and place LoRa relays at stops — a deterministic backbone instead of luck.
- Honest best-effort delivery model; SOS augments, never replaces, official emergency services.
- Hotel-verified fares plus post-ride traveller confirmation; AES-signed, rate-limited beacons.
- Adaptive duty-cycling and low-power BLE advertising; bulk sync only at the kiosk.
- CRDT conflict-free merge with tamper-evident sync across nodes.

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IMPACT AND BENEFITS

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Direct Impact on Target Users

- Tourists & solo travellers: confidence to explore off-grid, with fair fares and an offline safety net; language help to speak with locals.
- Hotels, homestays & eco-resorts: become the trusted network hub — a real reason for guests to book remote stays, raising footfall and revenue.
- Informal transport operators: shared jeep, bus and auto drivers gain visible, verified demand and dignified, transparent pricing.
- Women, families and first-time travellers: a safety net in areas they would otherwise avoid entirely.

Economic & Strategic Benefits

- Channels tourism spending directly into the rural economy — hospitality and informal transport.
- Requires zero new telecom infrastructure, so cost per additional village is negligible.
- Ends the “tourist tax”: verified fares can save a traveller ₹200+ on a single shared-jeep ride.
- Scales across states by adding regional map and fare packs — one app, every off-grid belt.
- Encourages low-impact, distributed tourism away from overcrowded hotspots.

Strategic Impact

- Builds an indigenous, offline safety and mobility layer for Indian tourism that depends on no foreign infrastructure.
- Directly supports Digital India, the Vibrant Villages Programme and Dekho Apna Desh.
- Advances Atmanirbhar Bharat by turning existing phones and public transport into national digital infrastructure.

Social & Environmental Benefits

- Safer travel for women, families and solo tourists in low-connectivity regions.
- Bridges language and culture between travellers and local communities.
- Spreads tourist load away from crowded honeypot destinations, reducing local environmental strain.

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Research and Analysis

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Gap & Problem Identification

- Large stretches of India’s hill, forest and border tourist belts remain network dead zones, leaving travellers without maps or a way to call for help.
- Existing offline maps (e.g. Google Maps offline) carry no informal transit routes, no verified fares and no peer-to-peer safety layer.
- Informal shared-jeep and auto fares are unpublished, so tourist overcharging is routine and undocumented.

Literature Survey & Benchmarking

- Delay-Tolerant Networking and the Bundle Protocol (RFC 5050) — store-and-forward messaging over intermittent links.
- DakNet (MIT Media Lab) and KioskNet — buses and kiosks proven as physical “data mules” bringing connectivity to rural villages.
- Haggle and pocket-switched networking research — opportunistic phone-to-phone message relay.
- Benchmark vs satellite SOS (iPhone 14+/Android): those need clear sky, premium handsets and paid plans; our mesh runs on cheap Android under tree cover.

Technology Validation

- Google Nearby Connections and Wi-Fi Aware are production Android APIs for offline peer-to-peer transport.
- Meshtastic / LoRa provides low-cost, kilometre-range off-grid mesh radio.
- On-device TTS (TensorFlow Lite, Piper) enables offline regional-language voice.
- CRDTs give conflict-free merge of offline-edited fare and route data.

Prototype Results & Policy Alignment

- Our working prototype demonstrates the full flow end-to-end: hotel-to-phone offline sync, offline fare lookup, and a multi-hop SOS relay reaching authorities.
- Demo runs with zero internet connectivity, validating the offline-first architecture.
- Aligns with Digital India, Vibrant Villages Programme, Dekho Apna Desh and Atmanirbhar Bharat.
- Note: SOS relay is a best-effort augmentation, not a replacement for official emergency services.